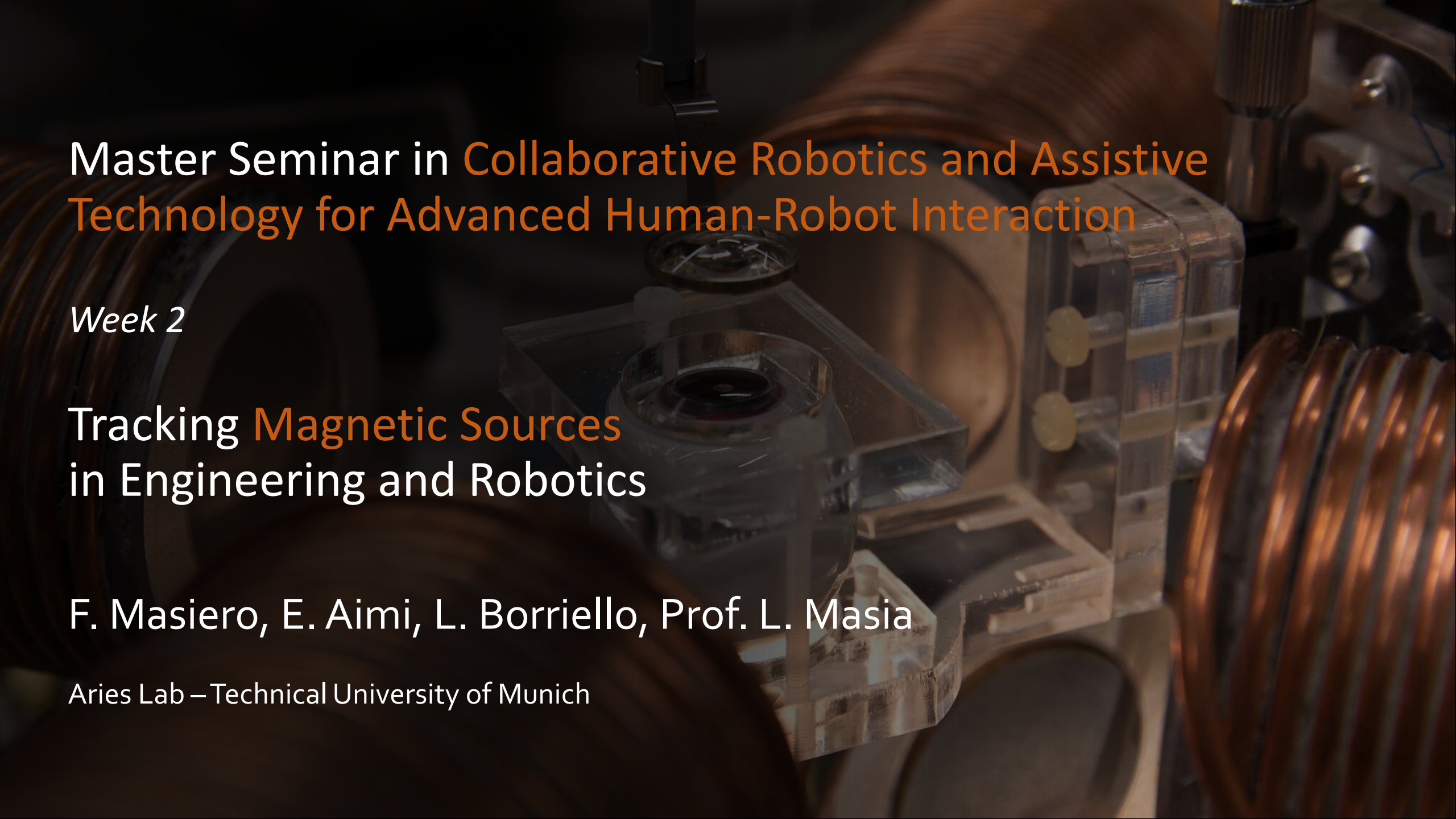




Master Seminar in Collaborative Robotics and Assistive Technology for Advanced Human-Robot Interaction

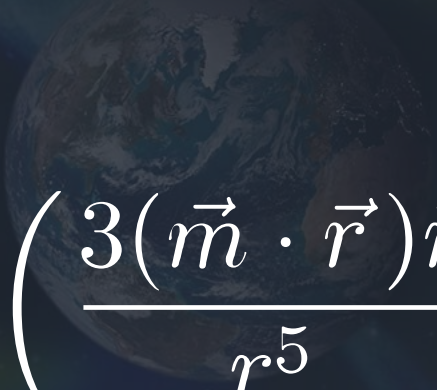
Week 2

Tracking Magnetic Sources in Engineering and Robotics

F. Masiero, E. Aimi, L. Borriello, Prof. L. Masia

Aries Lab – Technical University of Munich

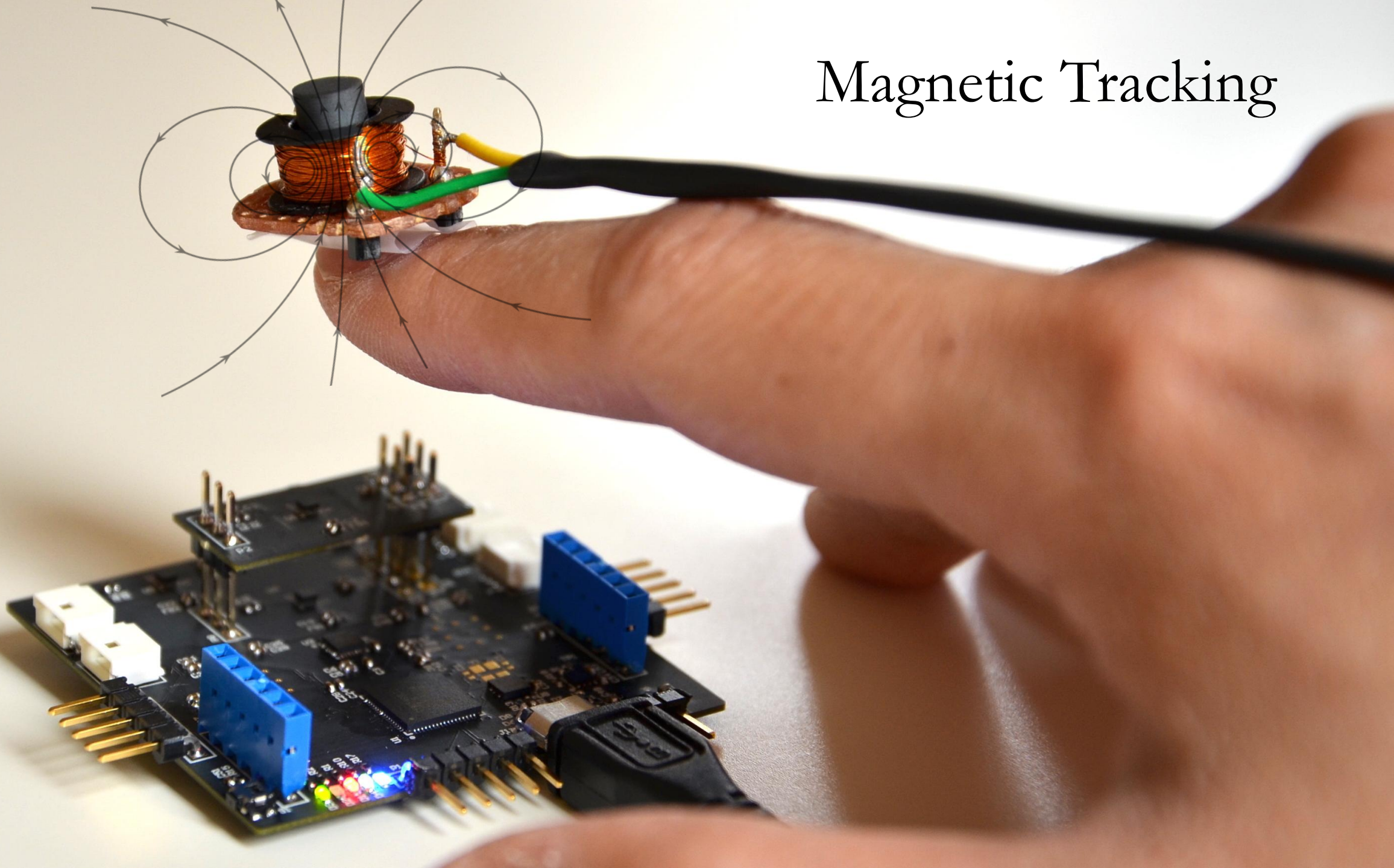
“As long as the observation point is sufficiently far away, the field of a body can be described as that of a magnetic dipole.”

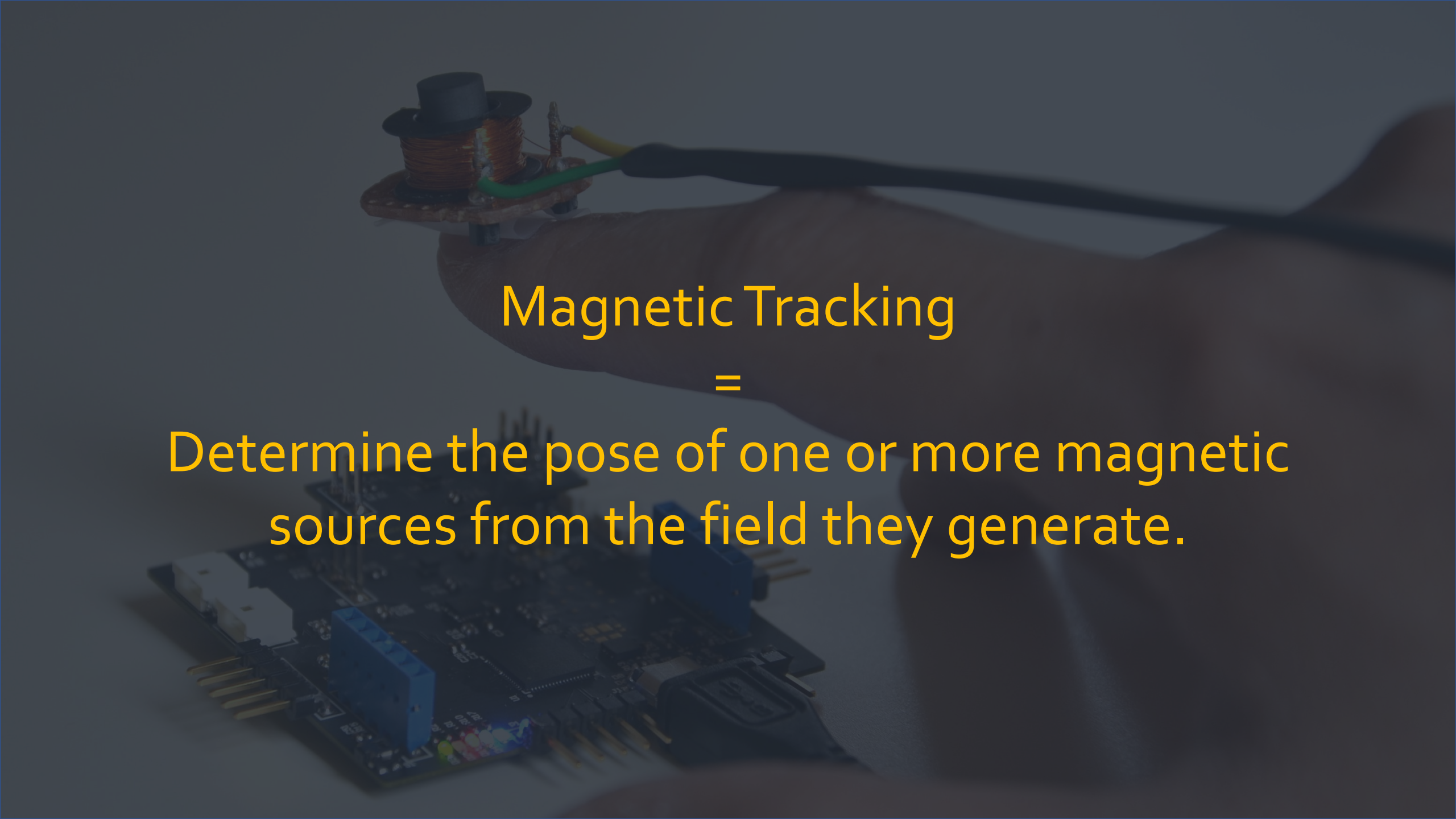

$$\vec{B} = \frac{\mu_0}{4\pi} \left(\frac{3(\vec{m} \cdot \vec{r})\vec{r}}{r^5} - \frac{\vec{m}}{r^3} \right)$$

$\vec{m}$: Magnetic Moment

$\vec{r}$: Distance from Magnetic Body

Magnetic Tracking



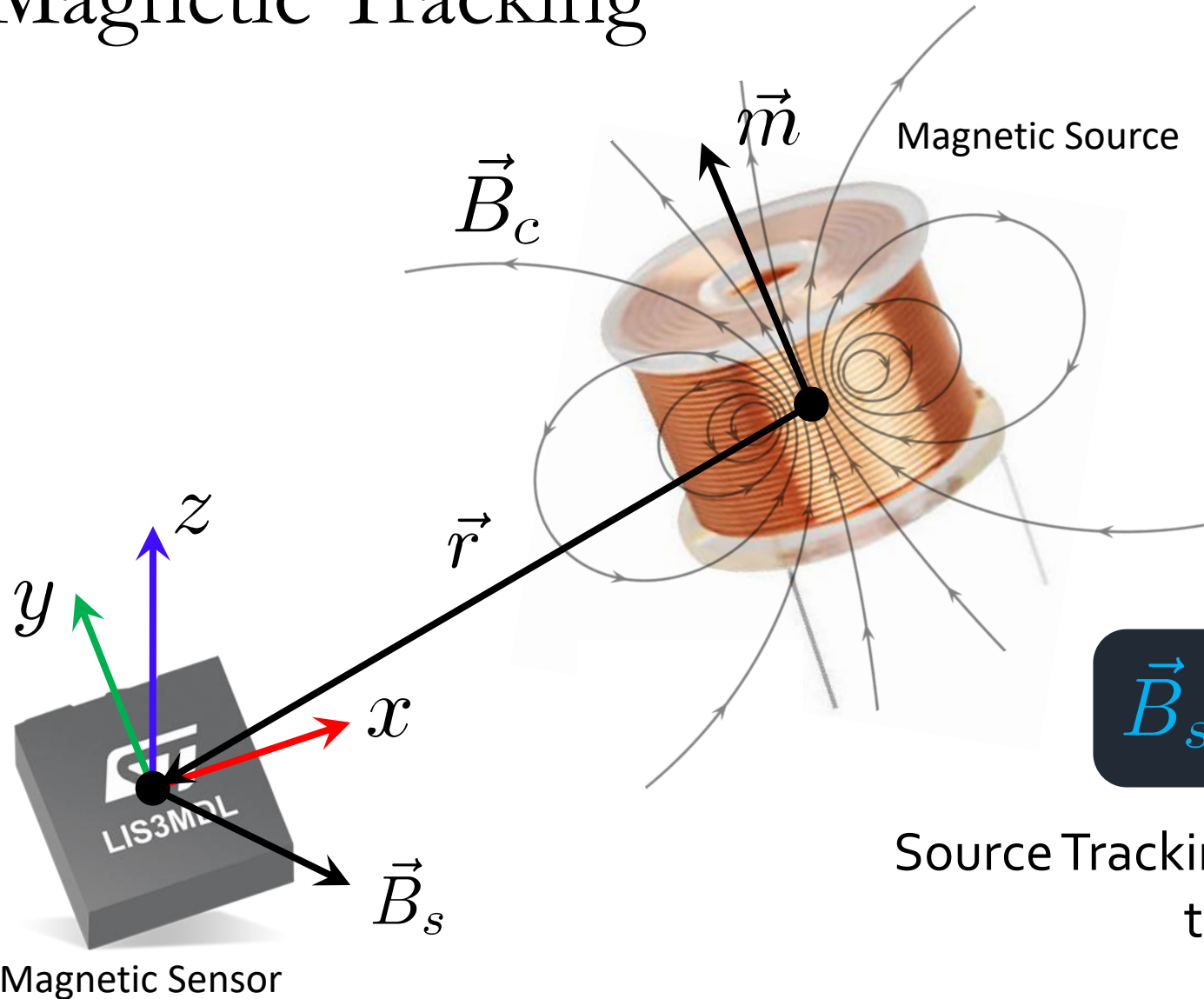
A hand is holding a small electronic component, which appears to be a coil or inductor, with a black cylindrical core and copper wire. The component is connected to a green and a yellow wire. In the background, a circuit board with various components, including blue connectors and a USB cable, is visible.

Magnetic Tracking

=

Determine the pose of one or more magnetic sources from the field they generate.

Magnetic Tracking



- Sensor
- Magnetic Source
- Geomagnetic Field
- Disturbance

$$\vec{B}_s = \vec{B}_c + \vec{B}_t + \vec{B}_d$$

Source Tracking can be obtained by inverting the equation above

Magnetic Tracking

The problem is **nonlinear** and should be solved numerically via optimization.

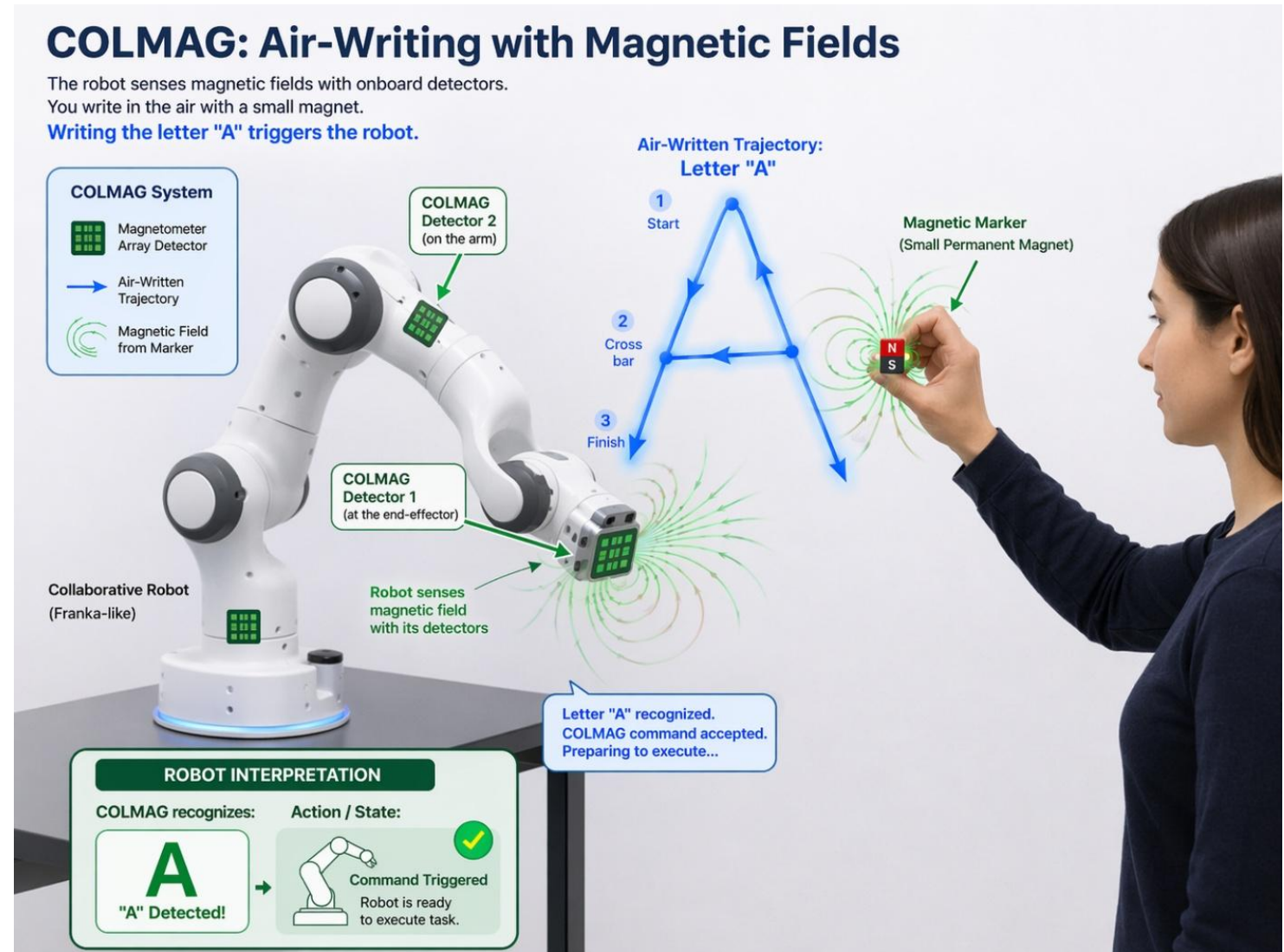
For example, if the sensors do not read external disturbances:

$$\vec{B}_s = \vec{B}_c \quad \longrightarrow \quad \min \left(\vec{B}_s - \vec{B}_c \right)$$

The algorithm looks for the **position** and **orientation** of the source by **minimizing the error** between **model** and **magnetic field readings**.

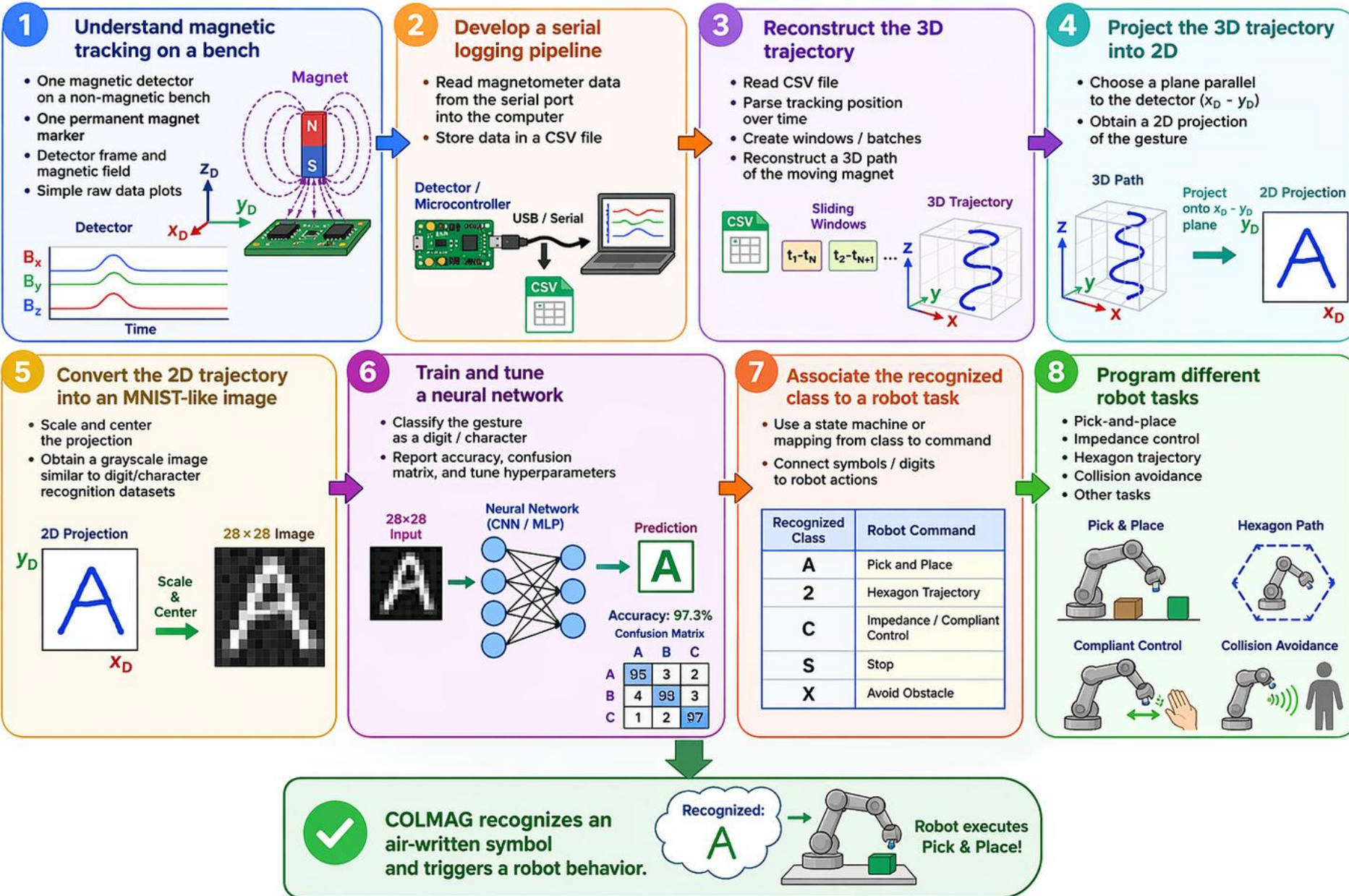
Projectual Work

Research project with an 8-week plan



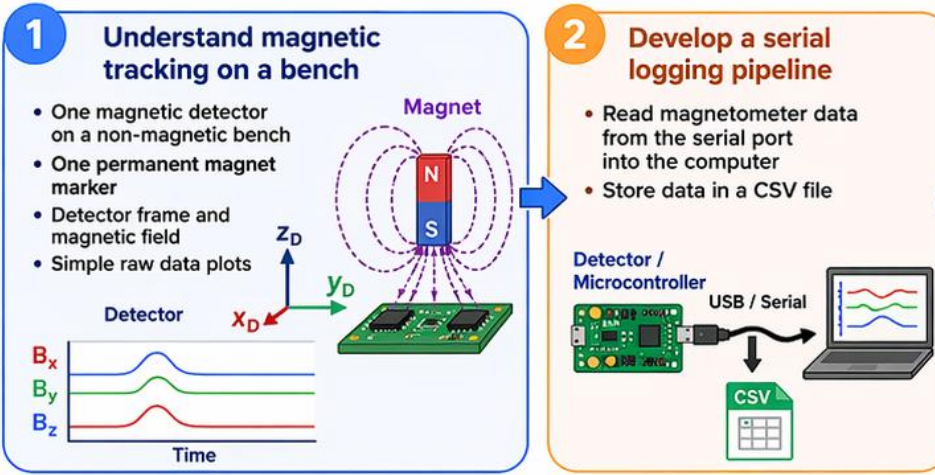
COLMAG Work Plan

From magnetic sensing to robot actions



COLMAG Work Plan

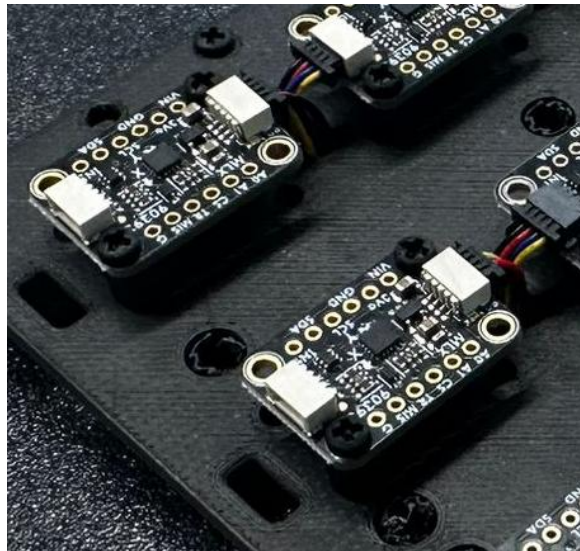
From magnetic sensing to robot actions



- We used an array of **16 three-axis magnetometers**, each capable of measuring the three spatial components of the magnetic field: x , y , and z .
- As a result, the array produces a data stream containing **3×16 magnetic field measurements**, for a total of **48 values** per sample.
- Using this detector, we can record the magnetic field generated by a permanent magnet. In this course, we will use **neodymium permanent magnets**.



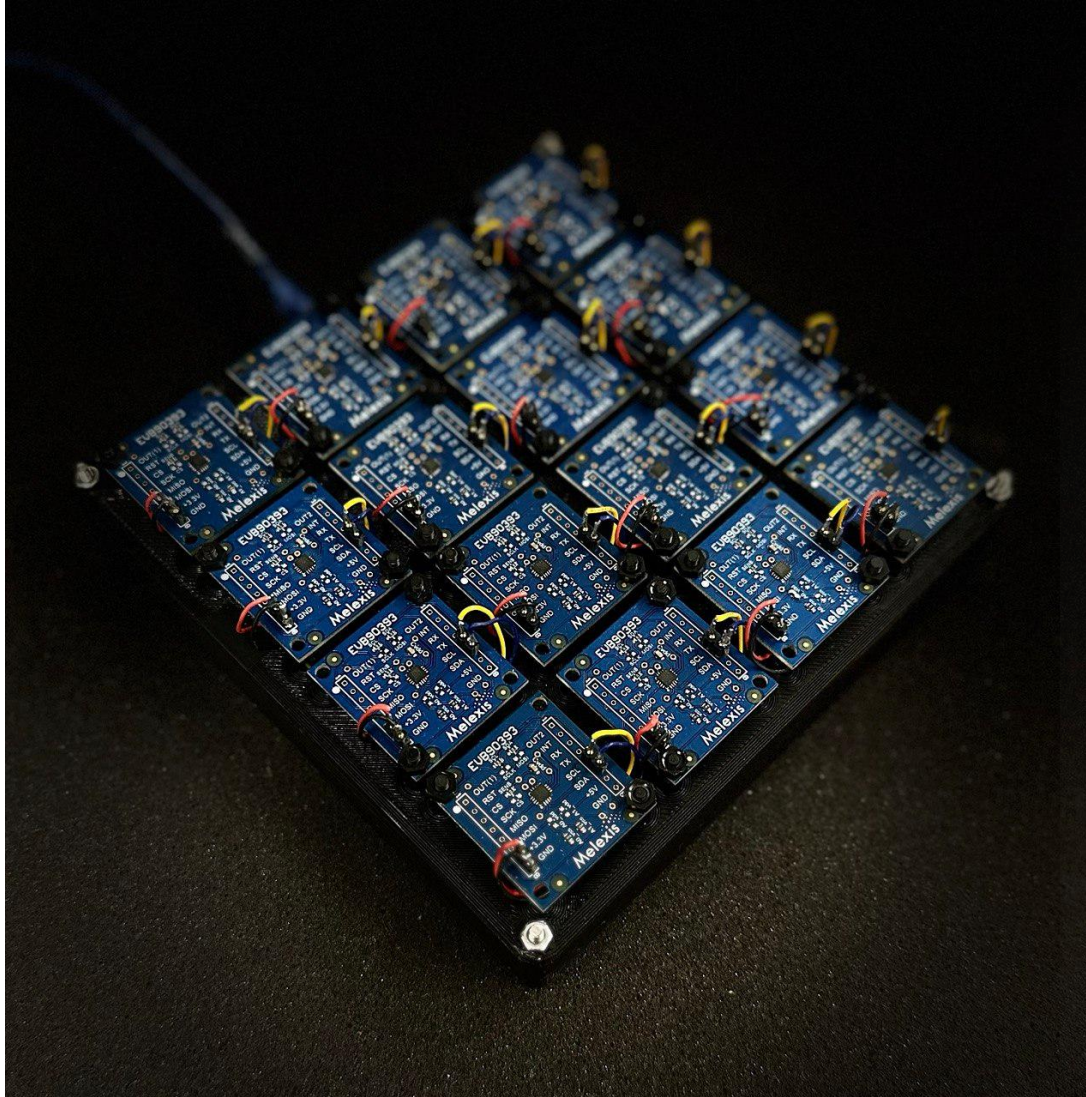
NdFeB magnet



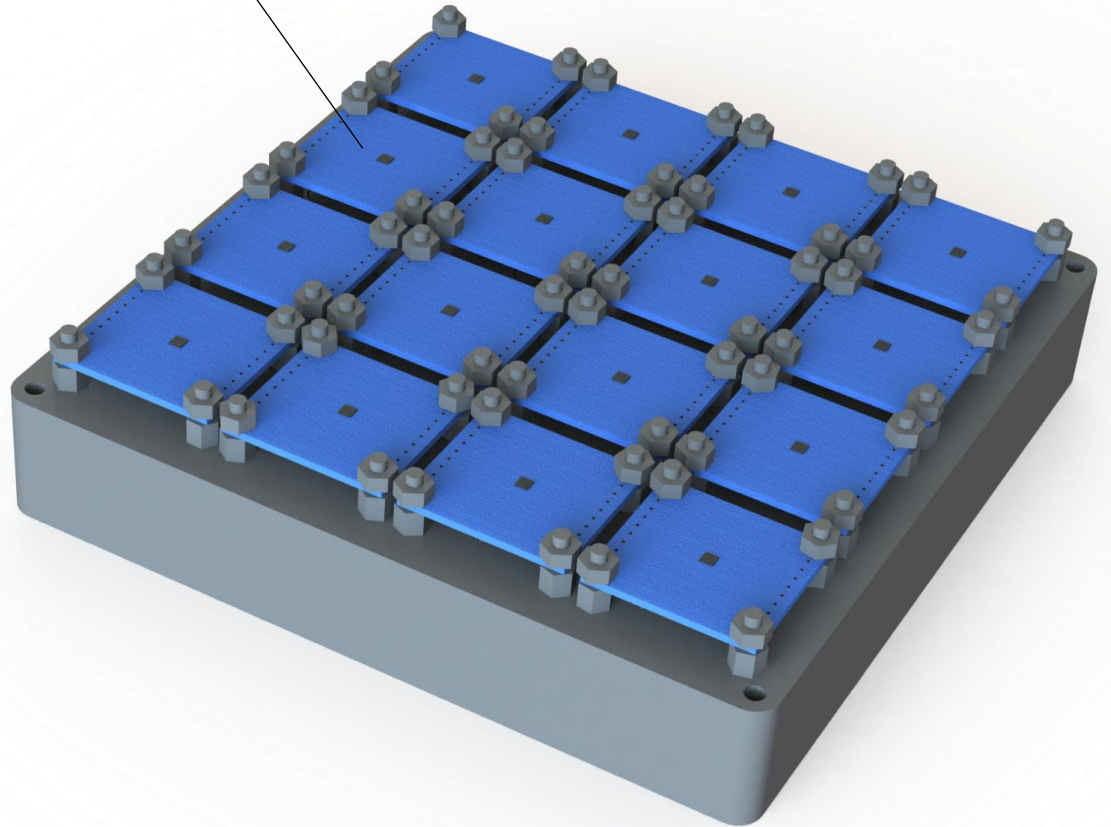
MLX90393, Melexis

<https://www.melexis.com/en/product/mlx90393/triaxis-micropower-magnetometer>

Projectual Work – Warmup Start

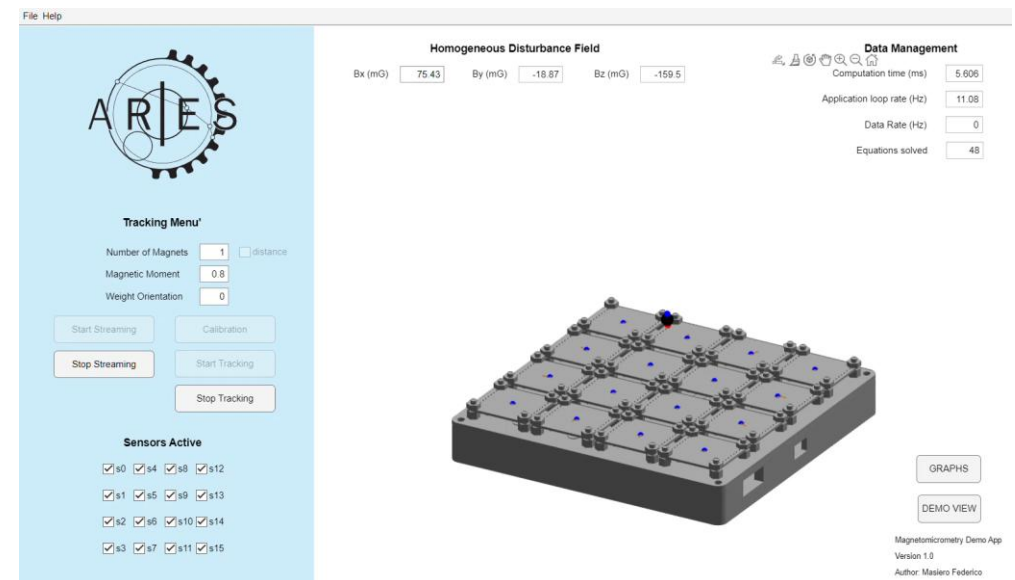
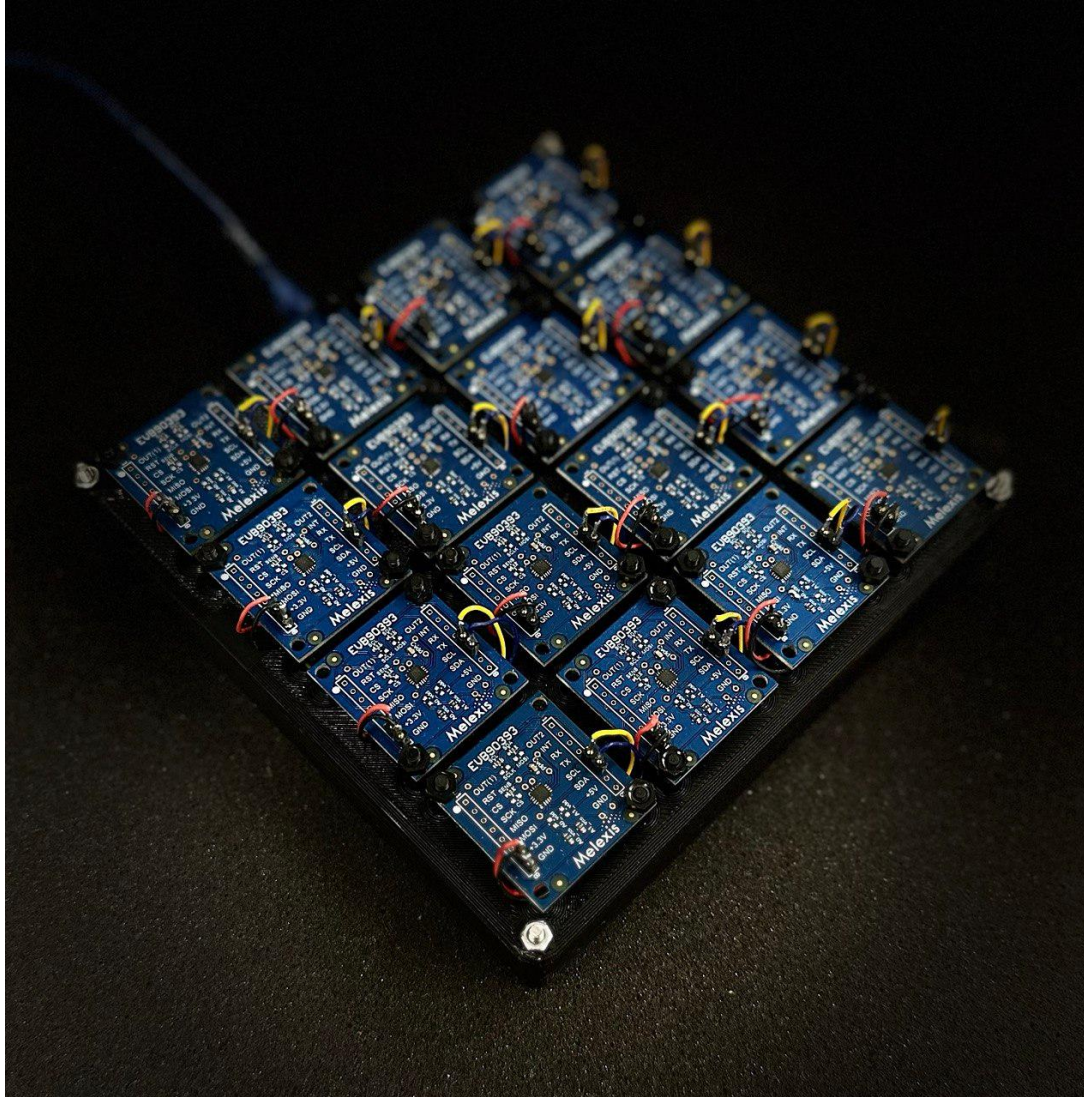


3D magnetometer

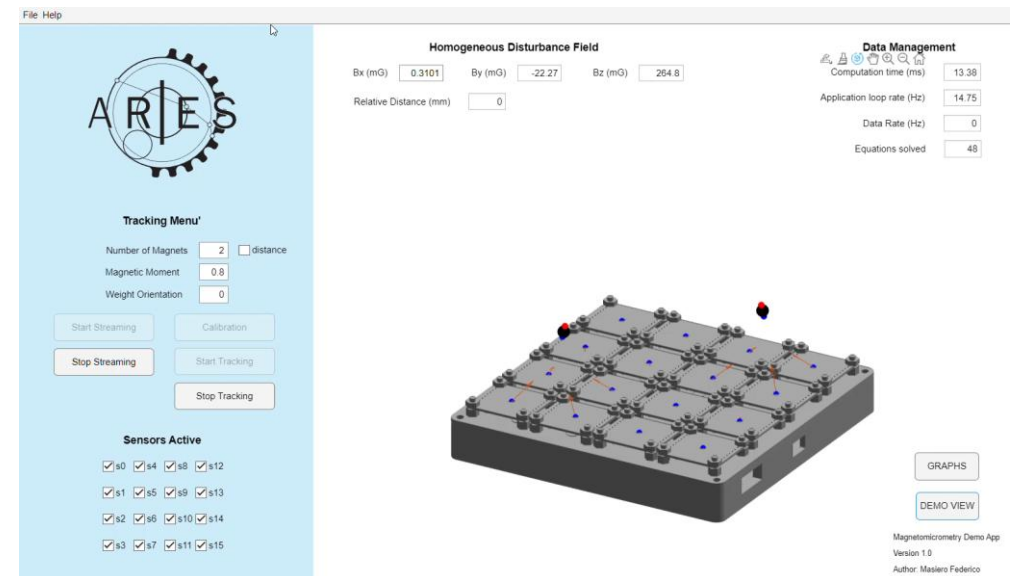
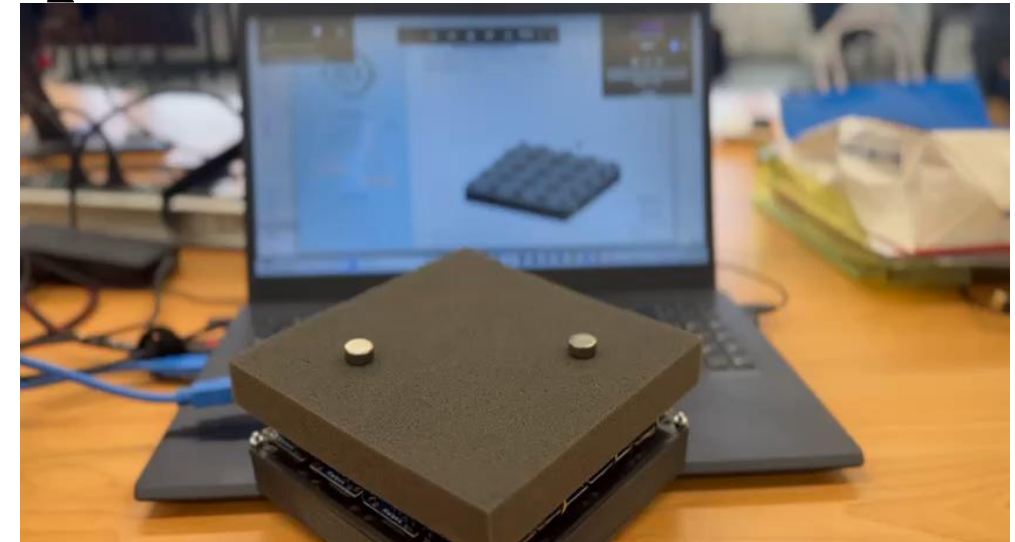
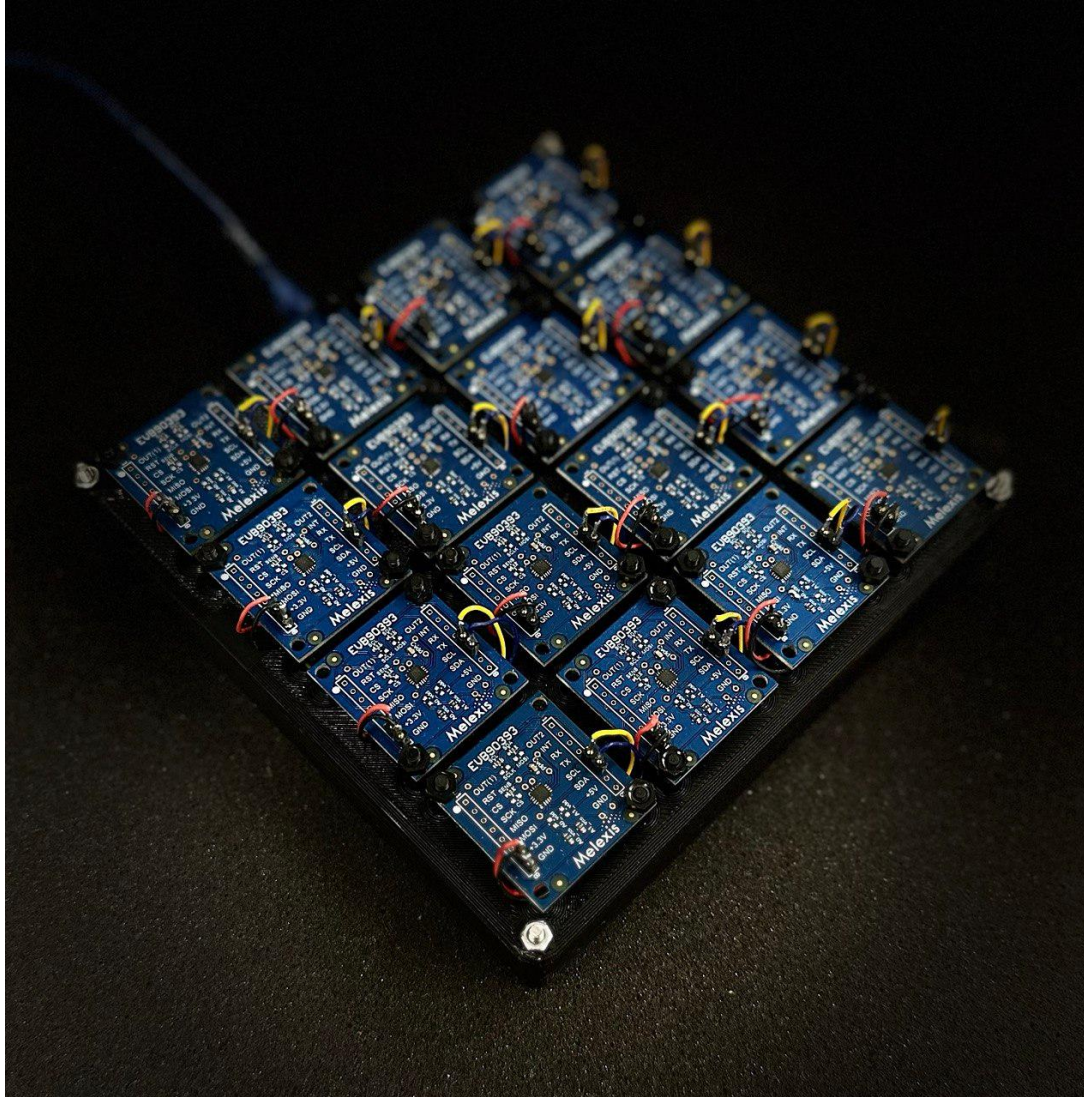


- We now check how the tracking looks like in our MATLAB app.

Projectual Work – Warmup Start



Projectual Work – Warmup Start



Projectual Work – Warmup Start

What is done in the App

$$\min \left(\vec{B}_s - \vec{B}_c \right)$$

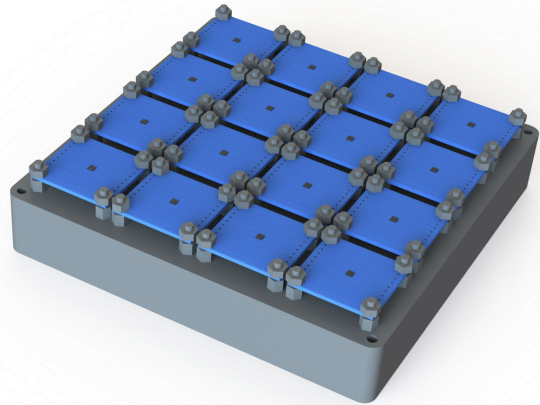
Magnet Tracking Pipeline



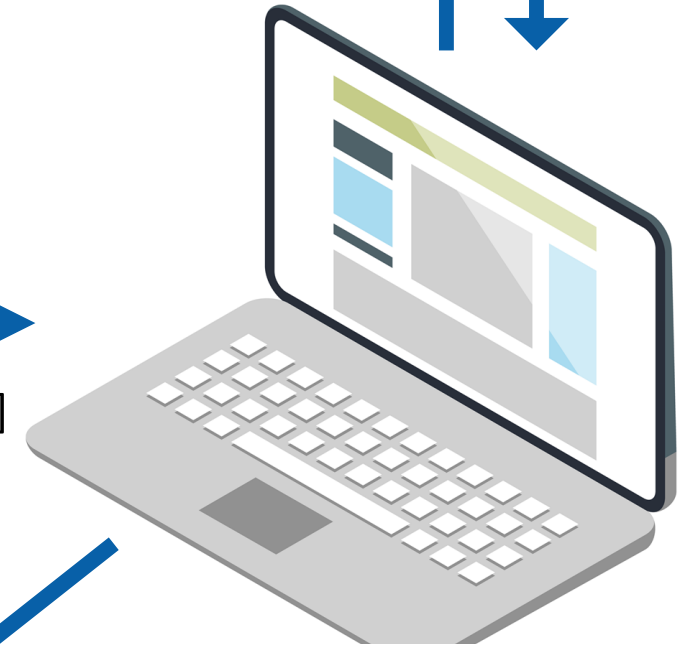
USB connection

Serial Communication

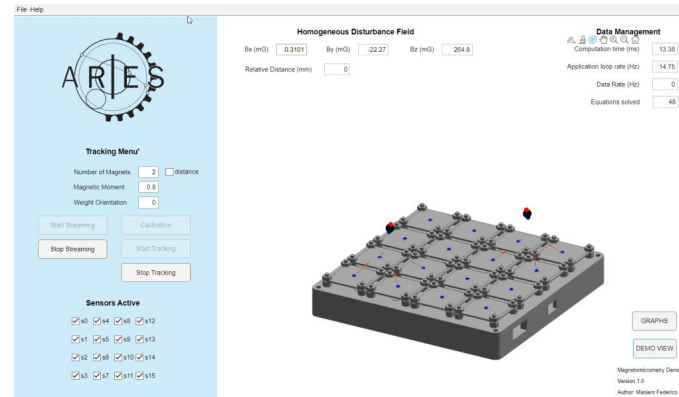
[Bx1, By1, Bz1, Bx2, By2, Bz2, ..., Bx16, By16, Bz16]



Magnetic Detector



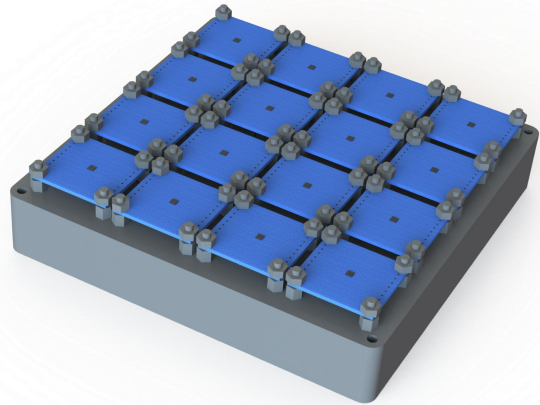
Laptop/Computer



Visualization

Projectual Work – Warmup Start

What is done in the App



Magnetic Detector

USB connection
Serial Communication

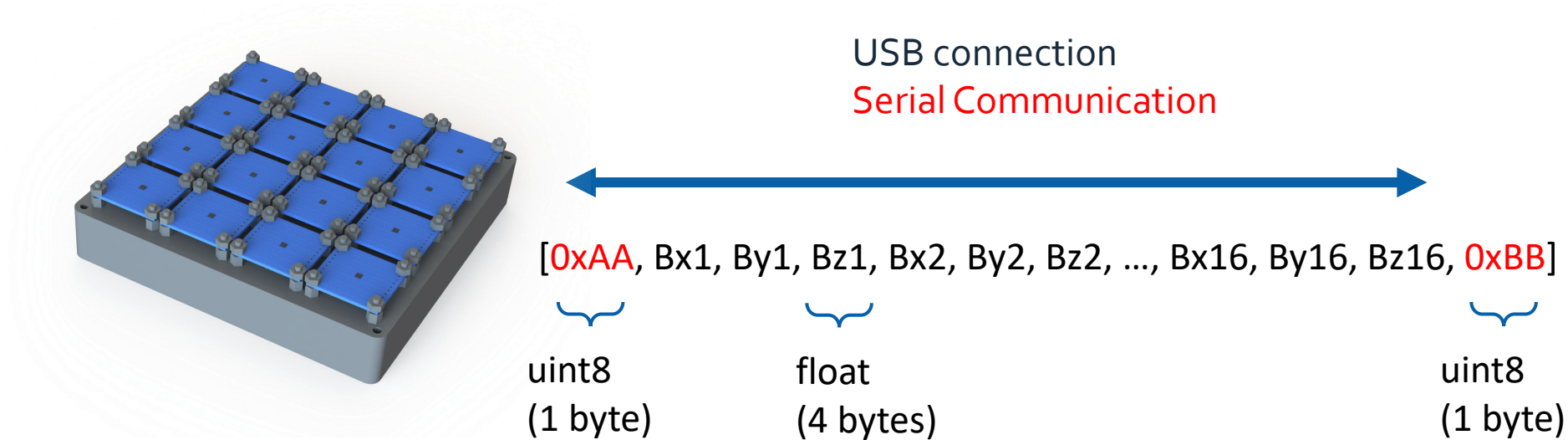


Basics of Serial Communications

- You need to know the protocol used by the device to communicate data, which includes:
 - Number of data being transmitted
 - Data type of the data being transmitted
 - If some flag characters are used in the protocol to detect the message correctly.
- Other information pertains some settings of the communication; mandatory parameters are
 - Baudrate (Bandwidth of the communication)
 - Port number (for the Laptop/computer only)

Projectual Work – Warmup Start

What is done in the App



Magnetic Detector

- A good data stream normally uses a Header (**0xAA**) and a Tail (**0xBB**) to have a sanity check of the data stream.
- How many bytes is this stream providing?

Projectual Work – Before Starting

- ☐ **Create a Branch** in the Lab Repo with your Name
- ☐ You Must work only on **YOUR BRANCH**
- ☐ The **main** branch is used and managed **SOLELY** by the instructors

Projectual Work – Your Task

GOAL: Read binary data from a microcontroller or embedded board through a serial port, validate each received data packet, and print the decoded values in the terminal. After that, optionally save the file in a CSV add live plotting using Matplotlib.

Start:

1. Install the required package:

```
</> Bash  
pip install pyserial
```

2. Connect the board to your laptop using USB
3. Identify the serial port used by the device

Operating system	Example port name
Windows	COM3 , COM4 , ...
Linux	/dev/ttyUSB0 , /dev/ttyACM0 , ...
macOS	/dev/tty.usbserial-* , /dev/tty.usbmodem-*

Projectual Work – Your Task

4. Set the Baudrate to 921600.
5. Each message sent by the board has the following structure:

Data packet format

Each message sent by the board has the following structure:

HEADER | DATA | TAIL

where:

HEADER = 0xAA
TAIL = 0xBB

The data section contains:

48 magnetic field values
6 tracking algorithm output values

Therefore, each packet contains:

$48 + 6 = 54$ float values

IMPORTANT NOTE: the magnet detection is done by the board in your project.

Note: float in the detector are 4 bytes numbers.

Then how many bytes should you read?

Projectual Work – Your Task

7. Create a Python Program that:

Main Script

- Lists the available serial ports.
- Let the user choose the correct port.
- Opens the serial port using: baudrate = 921600
- Creates a separate thread that reads the serial port periodically at 100 Hz.
- In the reading thread:
 - Read incoming bytes from the serial buffer.
 - Search for the header byte 0xAA.
 - Check that the tail byte 0xBB is in the expected position.
 - Confirm that the packet contains exactly 54 floats between the header and tail.
 - Decode the 54 floats (first 48 are mag field data, last 6 are mag pose).
 - Print the decoded data in the terminal.
- Safely close the serial port when the program exits.

Thread content

Projectual Work – Your Task

IMPROVEMENTS

7. Create a Python Program that:

Main Script

- Lists the available serial ports.
- Let the user choose the correct port.
- Opens the serial port using: baudrate = 921600
- Creates a separate thread that reads the serial port periodically at 100 Hz.
- In the reading thread:

Thread content

- Read incoming bytes from the serial buffer.
- Search for the header byte 0xAA.
- Check that the tail byte 0xBB is in the expected position.
- Confirm that the packet contains exactly 54 floats between the header and tail.
- Decode the 54 floats (first 48 are mag field data, last 6 are mag pose).
- Print the decoded data in the terminal.
- **Store Data in a CSV file row-wise format (Bx1 By1, By16, Bz16, x, y, z, mx, my, mz)**
- **Data recorded in the thread are saved in a dedicated variable and plotted using matplotlib (Mutex?)**
- Safely close the serial port when the program exits.

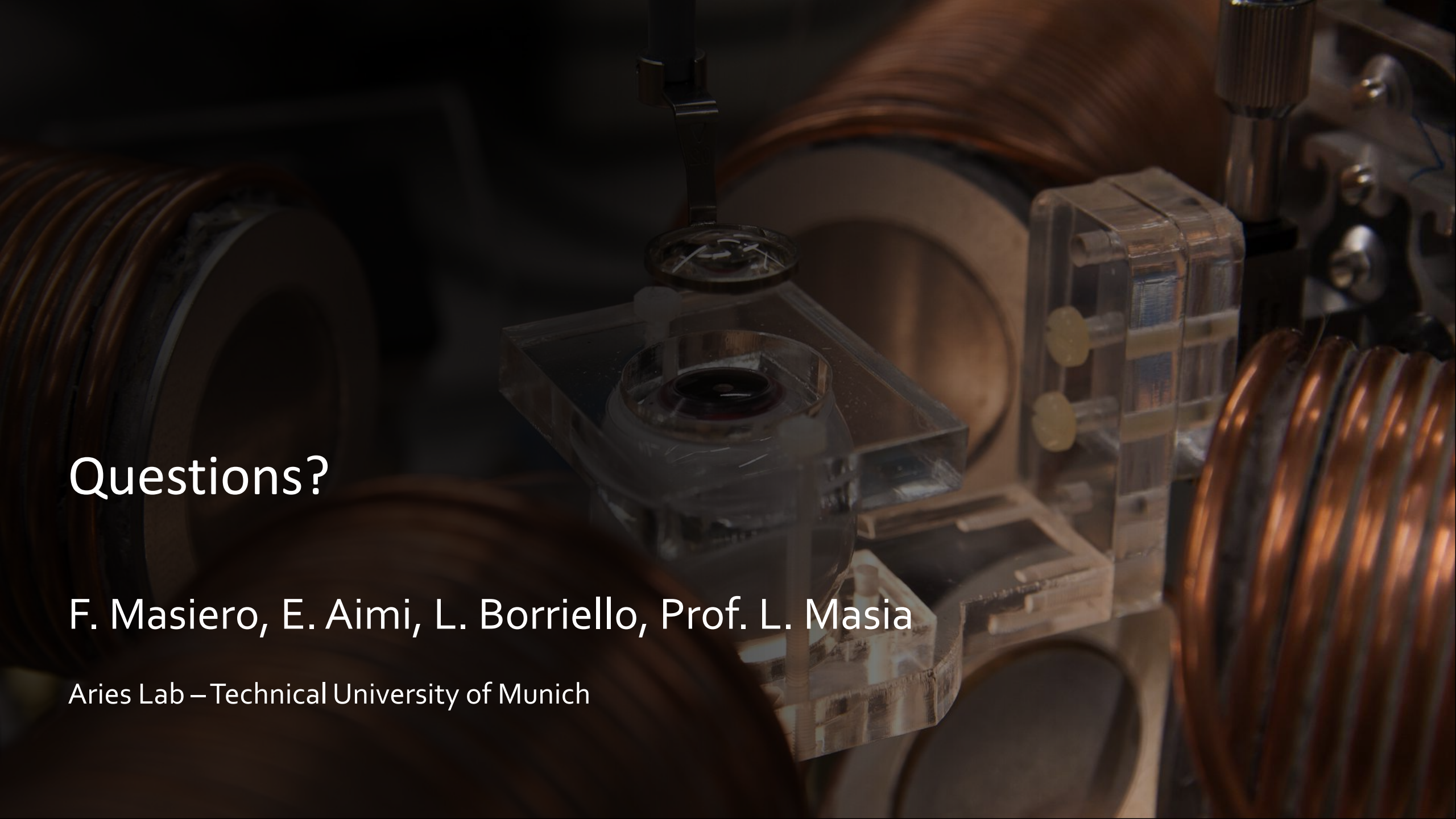
Projectual Work – Your Task

8. If you manage to do the CSV file writer, record three data series:
 - Without the magnet on top of the detector (does the detector recognize something?)
 - With a magnet still on top of the detector
 - With a magnet moving on top of the detector
9. Prepare a script to read and visualize the data from the CSV.

Projectual Work – Common Practice

THE HOLY RULES

- ☐ Do you think your solution is the best? Maybe not, discuss it with the others and with us; try to improve it!
- ☐ Keep your code tidy and well documented, not only in all the scripts you produce, but also in the repo Readme.MD
- ☐ Keep the repo clean and updated.



Questions?

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